Solutions: Problem Set 4

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Problem 4.1 (One-stage cluster sampling). The California Academic Performance Index (API) was a measure of school performance based on standardized tests. API scores ranged from 200 to 1,000, with higher values indicating better performance. The last API report was produced in 2013.

The API data are included in the `survey` package and can be loaded with `data(api)`. We use the one-stage cluster sample `apiclus1`. A simple random sample of $n = 15$ school districts was drawn from $N = 757$ districts, and all schools in the selected districts were observed.

The survey design is specified as follows:

```
dn_api <- svydesign(id = ~dnum, weights = ~pw, fpc = ~fpc, data = apiclus1)
```

Here, `dnum` identifies the school district, `pw` contains the design weights, and `fpc` contains the finite-population correction for sampling districts. We focus on `api00`, the API score in 2000.

a) Estimate the population mean API score in 2000. Also compute the design effect by calling `svymean()` with `deff = TRUE`.

Solution. Load the data and define the survey design:

```
> library("survey")
> data(api)
> dn_api <- svydesign(id = ~dnum, weights = ~pw, fpc = ~fpc, data = apiclus1)
```

The estimated mean and its design effect are:

```
> mean_api <- svymean(~api00, dn_api, deff = TRUE)
> mean_api

         mean     SE   DEff
api00 644.169 23.542 9.3459
```

The design effect is about 9.3. Thus, the 183 sampled schools contain roughly as much information about the population mean as an SRS of 20 schools. The loss of efficiency is caused mainly by the similarity of schools within the same district.

b) Estimate the 25th percentile, median, and 75th percentile of the API score in 2000. Use

```
svyquantile(x, design, quantiles, ...)
```

Solution. We estimate all three quartiles at once:

```
> q_api <- svyquantile(~api00, dn_api, quantiles = c(0.25, 0.5, 0.75))
> q_api

$api00
     quantile ci.2.5 ci.97.5       se
0.25      552    492     627 31.47166
0.5       652    561     714 35.66788
```

```
0.75      719    696     777 18.88300

attr(,"hasci")
[1] TRUE
attr(,"class")
[1] "newsvyquantile"
```

The estimated median is 652. Its 95 % confidence interval is 561 to 714.

c) The California State Board of Education used an API score of 800 as a performance target. Estimate the proportion of schools with an API score below 800. Use

svycdf(formula, design, ...)

Solution. `svycdf()` estimates the population cumulative distribution function:

```
> F_hat <- svycdf(~api00, dn_api)
```

Since API scores are integer-valued, the proportion below 800 is $\widehat{F}(799)$:

```
> p_below <- F_hat[[1]](799)
> p_below

[1] 0.9234973
```

Thus, about 92 % of schools have an API score below the target.

```
> plot(F_hat[[1]])
> abline(v = 800, lwd = 2)
```

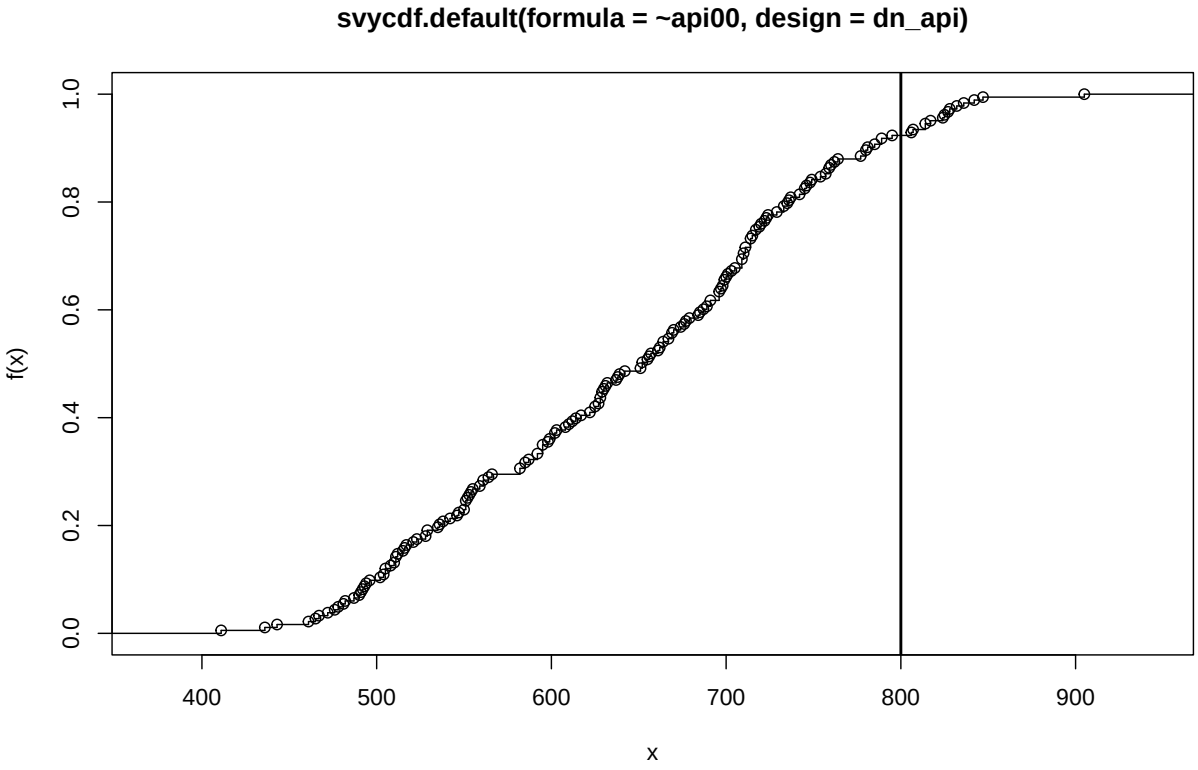
Problem 4.2 (Domain estimation). We use the data set `sepe.dat` from Problem Set 1. Treat the data as a simple random sample without replacement (SRS) of $n = 800$ establishments from a population of size $N = 3770$. We focus on industry class 31, that is, the domain U_{31} identified by `code == 1`.

a) Assume that the domain size N_{31} is unknown. Estimate the domain population mean $\bar{y}_{U31}$ of environmental-protection expenditures and its standard error.

> **Hint.** If `dn` is a survey design created with `svydesign()`, use `subset(dn, code == 1)` to restrict the design to the domain. Do not create a new SRS design from the domain observations: under the original SRS, the domain sample size n_{31} is random.

Solution. First load the data set `sepe.dat`, which is an SRS of size $n = 800$, and define the population size $N = 3770$:

```
> sepe <- read.table("https://stat.ethz.ch/Teaching/Datasets/cas-das/sepe.dat",
                     header = TRUE)
> sepe$I31 <- as.integer(sepe$code == 1)
> N <- 3770
```



Calculation with the `survey` package: First define the survey design object for the full sample:

```
> library("survey")
> sepe$fpc <- N
> dn_sepe <- svydesign(ids = ~1, fpc = ~fpc, data = sepe)
```

For domain estimation, it is important to distinguish between subsetting a survey design object and creating a new design object from the observations in the domain. The command

```
> dn_sepe_31 <- subset(dn_sepe, I31 == 1)
```

retains the original sampling design. Thus, establishments in industry class 31 are treated as a domain of the original SRS of $n = 800$ establishments. In particular, the number n_{31} of sampled domain units is random.

The domain population mean $\bar{y}_{U31}$ and its standard error are estimated by

```
> m_y_31 <- svymean(~expend, dn_sepe_31)
> m_y_31

          mean     SE
expend 113052 6432.8
```

By contrast, suppose that we first restrict the data to industry class 31 and then define a new design:

```
> sepe_31 <- subset(sepe, I31 == 1)
> dn_sepe_31_new <- svydesign(ids = ~1, data = sepe_31)
> svymean(~expend, dn_sepe_31_new)

          mean     SE
expend 113052 7269.4
```

This specifies a different sampling design. It treats the n_{31} observed establishments as if they formed a new equal-probability sample, rather than a domain of the original SRS. The estimated mean is the same because both estimators reduce to the sample mean within the domain. The estimated standard errors differ, however, because the two design objects represent different sampling designs.

Remarks. Let $\mathbb{1}_{i,31} = \mathbb{1}\{i \in U_{31}\}$ denote the domain indicator and let $s_{31} = s \cap U_{31}$. Then

$$n_{31} = \sum_{i \in s} \mathbb{1}_{i,31}.$$

The domain mean is estimated by

$$\bar{y}_{31} = \frac{\sum_{i \in s} \mathbb{1}_{i,31}\, y_i}{\sum_{i \in s} \mathbb{1}_{i,31}} = \frac{1}{n_{31}} \sum_{i \in s_{31}} y_i.$$

Because n_{31} is random under the original SRS, this estimator can be viewed as a ratio estimator. A linearization variance estimator under SRS is

$$\widehat{\mathrm{var}}_p\left(\bar{y}_{31}\right) = (1-f)\frac{n(n_{31}-1)}{n_{31}^2(n-1)}\,\widehat{\sigma}^2_{y,31}, \qquad f = \frac{n}{N},$$

where $\widehat{\sigma}^2_{y,31}$ is the sample variance of y among the n_{31} sampled domain units.

Calculation with R:

```
> n31 <- sum(sepe$I31)
> mean_y_31 <- mean(sepe$expend[sepe$I31 == 1])
> sigma2_y_31 <- var(sepe$expend[sepe$I31 == 1])
> se_mean_y_31 <- sqrt((1 - n / N) * n * (n31 - 1) /
                       (n31^2 * (n - 1)) * sigma2_y_31)
```

The manually calculated quantities agree with those obtained from the subsetted survey design.

```
> identical(as.numeric(coef(m_y_31)), mean_y_31)

[1] TRUE

> all.equal(as.numeric(SE(m_y_31)), se_mean_y_31)

[1] TRUE
```

b) Now assume that the domain size $N_{31} = 650$ is known, but that this information was not used when the sample was drawn. Estimate the population domain total of environmental expenditures $t_{y,31}$ and its standard error.

Solution. The sampling design remains an SRS from the full population. Hence, n_{31} remains random. Following the classical definition of the hypergeometric distribution,[1] consider a population of N units, of which N_{31} belong to domain U_{31} and $N - N_{31}$ do not. If an SRS of n units is drawn without replacement, then the number n_{31} of sampled domain units follows a hypergeometric distribution:

$$n_{31} \sim \text{Hypergeometric}(N, N_{31}, n). \tag{1}$$

The known domain size $N_{31} = 650$ is used only at the estimation stage. The domain total is estimated by

$$\widehat{t}_{y,31} = N_{31} \cdot \bar{y}_{31},$$

with estimated standard error

$$\widehat{\mathsf{SE}}_p(\widehat{t}_{y,31}) = N_{31} \cdot \widehat{\mathsf{SE}}_p(\bar{y}_{31}).$$

The variance estimator for $\bar{y}_{31}$, and hence $\widehat{\mathsf{SE}}_p(\bar{y}_{31})$, is the linearization variance estimator under SRS introduced in part a).

Calculation with the `survey` package: We use the subsetted survey design `dn_sepe_31` from part a). The estimated domain mean $\bar{y}_{31}$ has already been calculated; it is reproduced here for ease of reference:

```
> m_y_31 <- svymean(~expend, dn_sepe_31)
```

The estimated domain total and its standard error are then

```
> N31 <- 650
> c(total = N31 * coef(m_y_31), SE = N31 * SE(m_y_31))
```

```
total.expend           SE
    73483508      4181303
```

Remarks. If the domain size N_{31} is known, then n_{31} has a hypergeometric distribution; see (1). In our case, n_{31} has expected value ≈ 137.9 and standard deviation ≈ 9.6. Thus, the observed value $n_{31} = 138$ is close to its expected value under the sampling design. For example, the central 80% range of the distribution can be obtained with:

```
> qhyper(c(0.1, 0.9), N31, N - N31, n)
```

```
[1] 126 150
```

Note that R uses an equivalent but different parametrization from that in (1).

Another remark is in order. The known domain size N_{31} can also be incorporated by calibration. We can define the domain indicator as an auxiliary variable and calibrate the full design to the known totals N and N_{31}:

```
> dn_sepe_cal <- calibrate(dn_sepe, ~I31,
                           population = c(`(Intercept)` = N, I31 = N31))
> svytotal(~expend, subset(dn_sepe_cal, I31 == 1))
```

[1] See e.g., Johnson, N. L, Kemp, A. W., and Kotz, S. (2005). Univariate Discrete Distributions, 3rd ed., Hoboken (NJ): John Wiley & Sons, p. 251.

```
           total      SE
expend 73483508 4181303
```

We will study calibration in more detail later.

Problem 4.3 (Stratified simple random sampling). We again use the data set `env.dat`. Assume that membership in industry class 31 was known from the sampling frame and that the population was stratified by this variable before sampling. Within each stratum, a simple random sample without replacement was drawn. The population consists of two strata:

Stratum	Population size
U_{31} (`code == 1`)	$N_{31} = 460$
U_{rest} (`code == 0`)	$N_{\text{rest}} = 3{,}310$

Hint. The stratum sample sizes are available in `sepe.dat`. As explained in Problem Set 2, create a survey design object for stratified sampling and name it `dn_sepe`.

a) Estimate the population total of environmental-protection expenditures using `svytotal()`. Also report a 90 % confidence interval using

```
confint(object, parm, level = 0.95, ...)
```

Solution. Load the data and define the survey design:

```
> library("survey")
> sepe <- read.table("https://stat.ethz.ch/Teaching/Datasets/cas-das/sepe.dat",
                     header = TRUE)

> sepe$Nph <- ifelse(sepe$code == 1, 460, 3310)
> dn_sepe <- svydesign(ids = ~1, strata = ~code, fpc = ~Nph, data = sepe)
```

The estimated total and its 90 % confidence interval are

```
> t_y <- svytotal(~expend, dn_sepe)
> t_y

           total      SE
expend 317824493 7719366

> confint(t_y, level = 0.90)

              5 %      95 %
expend 305127266 330521720
```

b) Estimate the mean environmental-protection expenditures in each stratum using `svyby()`.

Hint. The function `svyby(formula, by, design, FUN, ...)` applies the survey statistic `FUN` to the variables specified by `formula`, separately for the groups defined by `by`, using the survey design `design`. Additional arguments can be passed through `...`.

Next, test whether the two stratum means are equal. For comparison, use both

```
t.test(expend ~ code, data = sepe, var.equal = TRUE)
```

and the design-based test

```
svyttest(expend ~ code, dn_sepe)
```

The first is the classical pooled two-sample t-test and ignores the sampling design. The second uses the variance implied by the survey design, including the finite population corrections.

Solution. The estimated stratum means are

```
> svyby(~expend, ~code, dn_sepe, svymean)

  code    expend       se
0    0  80308.39 2173.576
1    1 113051.55 6082.014
```

The classical pooled two-sample t-test is

```
> t.test(expend ~ code, data = sepe, var.equal = TRUE)

        Two Sample t-test

data:  expend by code
t = -5.2217, df = 798, p-value = 2.264e-07
alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
95 percent confidence interval:
 -45052.07 -20434.25
sample estimates:
mean in group 0 mean in group 1
       80308.39       113051.55
```

The corresponding design-based test is

```
> svyttest(expend ~ code, dn_sepe)

        Design-based t-test

data:  expend ~ code
t = 5.0696, df = 797, p-value = 4.96e-07
alternative hypothesis: true difference in mean is not equal to 0
95 percent confidence interval:
 20065.01 45421.31
sample estimates:
difference in mean
          32743.16
```

The two tests need not give the same standard error or p-value. `t.test()` treats the observations as independent samples from two populations and pools their variances, whereas `svyttest()` uses a design-based variance estimate from `dn_sepe`.

c) Let y_i denote environmental-protection expenditures (`expend`) and x_i the number of full-time employees (`empl`). Two different summaries are commonly considered:

$$r_{AR} = \frac{1}{N_+} \sum_{i \in U_+} \frac{y_i}{x_i}, \qquad r_{RA} = \frac{\sum_{i \in U} y_i}{\sum_{i \in U} x_i},$$

where $U_+ = \{i \in U : x_i > 0\}$ and $N_+ = |U_+|$. The first is the average of ratios; the second is the ratio of totals (equivalently, the ratio of averages).

Estimate r_{RA} using `svyratio()` and the design object `dn_sepe`. Then repeat the calculation with `separate = TRUE`. What changes?

> **Hint.** Use
>
> *svyratio(numerator = formula, denominator, design, ...)*
>
> The default estimates one combined ratio. With `separate = TRUE`, ratios are estimated separately within the strata.

Solution. The combined ratio estimator is

$$\widehat{r}_{RA} = \frac{\widehat{t}_y}{\widehat{t}_x}.$$

```
> r_RA <- svyratio(~expend, ~empl, dn_sepe)
> r_RA

Ratio estimator: svyratio.survey.design2(~expend, ~empl, dn_sepe)
Ratios=
           empl
expend 450.1731
SEs=
           empl
expend 10.85506
```

Thus, the estimated ratio is 450.2, with estimated standard error 10.9. With `separate = TRUE`, `svyratio()` instead reports one ratio for each stratum:

```
> r_RA_sep <- svyratio(~expend, ~empl, dn_sepe, separate = TRUE)
> r_RA_sep

Stratified ratio estimate: svyratio.survey.design2(~expend, ~empl, dn_sepe, separate = TRUE)
Ratio estimator: Stratum == 0L
Ratios=
           empl
expend 461.3302
SEs=
```

```
           empl
expend 12.77544
Ratio estimator: Stratum == 1L
Ratios=
           empl
expend 400.6449
SEs=
           empl
expend 18.37954
```

Hence, `separate = TRUE` does not produce a second single estimate of the overall ratio; it produces the within-stratum ratios $\widehat{t}_{yh}/\widehat{t}_{xh}$.

d) Convert `dn_sepe` to a replicate-weight design for the Rao–Wu $(n-1)$ bootstrap with $B = 200$ replicates:

```
set.seed(11)
dn_sepe_boot <- as.svrepdesign(dn_sepe, type = "subbootstrap",
                               replicates = 200)
```

Re-estimate the combined ratio r_{RA} from part c) using this replicate design. Also report its 95 % confidence interval.

Solution. Create the replicate-weight design:

```
> set.seed(11)
> dn_sepe_boot <- as.svrepdesign(dn_sepe, type = "subbootstrap",
                                 replicates = 200)
```

Estimate the ratio:

```
> r_RA_boot <- svyratio(~expend, ~empl, dn_sepe_boot)
> r_RA_boot

Ratio estimator: svyratio.svyrep.design(~expend, ~empl, dn_sepe_boot)
Ratios=
           empl
expend 450.1731
SEs=
         [,1]
[1,] 12.14926
```

The point estimate is the same ratio estimator as before, while its standard error is now estimated from the bootstrap replicate weights. The 95 % confidence interval is

```
> confint(r_RA_boot)

                2.5 %   97.5 %
expend/empl 426.361 473.9853
```